DOC 2 — SOLUTIONS: EVIDENCE — DATA / GRAPH / TABLE — 50 MCQs

EASY (Q1–Q10)

Q1. Correct Answer: C

REASONING: The question asks for fluorite's hardness on the Mohs scale. Locate "Fluorite" in the table → the corresponding value is 4.

WHY EACH OPTION IS WRONG: - A) 1 is the hardness of talc, not fluorite. - B) 2 is the hardness of gypsum. - D) 7 is the hardness of quartz.

TRAP ALERT: Distractors are real values from the table but matched to the wrong mineral. Always trace your finger across the correct row.

💡 **KEY INSIGHT:** Easy lookup questions test whether you read the correct row, not whether you understand the science.

⚡ **FAST METHOD:** Find the keyword in the stem ("fluorite"), match it to the left column, read across.

Q2. Correct Answer: C

REASONING: The question asks for the average wingspan of the common starling. Locate "Common starling" → 38 cm.

WHY EACH OPTION IS WRONG: - A) 24 is the house sparrow's wingspan. - B) 33 is the barn swallow's wingspan. - D) 54 is the Eurasian jay's wingspan.

TRAP ALERT: The species are listed in descending order of wingspan, so the answer is neither the first nor the last entry. Don't assume position = answer.

💡 **KEY INSIGHT:** "According to the table" signals a pure lookup—no inference needed.

⚡ **FAST METHOD:** Scan the species column for "common starling," read the number in the same row.

Q3. Correct Answer: C

REASONING: Locate Portugal in the table → 26.4%.

WHY EACH OPTION IS WRONG: - A) 55.2% belongs to Denmark. - B) 33.5% belongs to Ireland. - D) 22.1% belongs to Germany.

TRAP ALERT: Each distractor is a real percentage from the table matched to the wrong country.

💡 **KEY INSIGHT:** When the stem names a specific entity, lock onto that row before looking at options.

⚡ **FAST METHOD:** Country → Portugal → 26.4 → answer C.

Q4. Correct Answer: C

REASONING: Compare the "Estimated population" column: 340, 10,000, 84,000, 27,000. The largest value is 84,000 (humpback whale).

WHY EACH OPTION IS WRONG: - A) 340 is the smallest population, not the largest. - B) 10,000 is the second-smallest population. - D) 27,000 is the second-largest population.

TRAP ALERT: Option D (gray whale, 27,000) is a plausible distractor because the number is large, but it is not the largest.

💡 **KEY INSIGHT:** "Largest" requires scanning the entire column before choosing.

⚡ **FAST METHOD:** Scan the rightmost column for the biggest number → 84,000 → humpback → C.

Q5. Correct Answer: D

REASONING: Locate "Gran Telescopio Canarias" → 10.4 meters.

WHY EACH OPTION IS WRONG: - A) 5.1 m belongs to Hale. - B) 8.4 m belongs to the Vera C. Rubin Observatory. - C) 10.0 m belongs to Keck I.

🚨 **TRAP ALERT:** C (10.0) and D (10.4) are very close values. Read the row carefully—Keck I ≠ Gran Telescopio Canarias.

💡 **KEY INSIGHT:** When two values are similar, precision in matching the name to the row is critical.

⚡ **FAST METHOD:** Find "Gran Telescopio Canarias" → read across to "Primary mirror diameter" → 10.4.

Q6. Correct Answer: D

REASONING: Compare January temperatures: −7, 16, 28, 33. The highest is 33°F (Sitka).

WHY EACH OPTION IS WRONG: - A) −7°F is the lowest (coldest), not the highest. - B) 16°F is the second-lowest. - C) 28°F is the second-highest.

🚨 **TRAP ALERT:** "Highest January temperature" in Alaska may feel counterintuitive, but the question is about the table values, not your expectations. Also, don't confuse the January and July columns.

💡 **KEY INSIGHT:** The table has two temperature columns. The stem specifies "January"—ignore July.

⚡ **FAST METHOD:** Focus on the January column only → scan for the largest number → 33 → Sitka → D.

Q7. Correct Answer: C

REASONING: The stem asks which nutrient spinach has MORE of per 100 g than kale. Compare each row: Vitamin C: 28 < 93 (kale wins). Calcium: 99 < 254 (kale wins). Iron: 2.7 > 1.6 (spinach wins). Fiber: 2.2 < 4.1 (kale wins). Only iron is higher in spinach.

WHY EACH OPTION IS WRONG: - A) Spinach has less vitamin C than kale (28 vs. 93), not more. - B) Spinach has less calcium than kale (99 vs. 254). - D) Spinach has less fiber than kale (2.2 vs. 4.1).

🚨 **TRAP ALERT:** The question asks where spinach exceeds kale, but options A, B, and D all show values where kale exceeds spinach. Read the comparison direction carefully.

💡 **KEY INSIGHT:** Each option states real data, but only one correctly identifies where spinach > kale.

⚡ **FAST METHOD:** For each nutrient, check: is spinach's value > kale's value? Only iron qualifies.

Q8. Correct Answer: C

REASONING: Compare elevations: 1,397; 1,050; 1,467; 887. The highest is 1,467 m (La Grande Soufrière).

WHY EACH OPTION IS WRONG: - A) 1,397 m is the second-highest. - B) 1,050 m is the third-highest. - D) 887 m is the lowest.

🚨 **TRAP ALERT:** Mount Pelée (1,397 m) is close to La Grande Soufrière (1,467 m). Don't rush the comparison.

💡 **KEY INSIGHT:** "Highest" means scan the entire column, not just stop at the first large number.

⚡ **FAST METHOD:** Scan elevation column → 1,467 is the max → La Grande Soufrière → C.

Q9. Correct Answer: D

REASONING: Compare thermal conductivity: 385, 205, 16, 1.0. The lowest is 1.0 W/m·K (glass).

WHY EACH OPTION IS WRONG: - A) Copper has the highest thermal conductivity, not the lowest. - B) Aluminum has the second-highest. - C) Stainless steel has the second-lowest (16), not the lowest.

🚨 **TRAP ALERT:** C (stainless steel at 16) is tempting because 16 is much lower than 385 or 205, but glass at 1.0 is even lower.

💡 **KEY INSIGHT:** Don't stop scanning after finding a small-looking number. Check every row.

⚡ **FAST METHOD:** Scan the rightmost column → 1.0 is clearly smallest → glass → D.

Q10. Correct Answer: C

REASONING: The stem asks about "the rover that landed in 2012." Locate 2012 → Curiosity → 899 kg.

WHY EACH OPTION IS WRONG: - A) 11.5 kg is Sojourner (1997). - B) 185 kg is Spirit (2004). - D) 1,025 kg is Perseverance (2021).

🚨 **TRAP ALERT:** The stem references landing year rather than rover name. You must match 2012 first, then read across.

💡 **KEY INSIGHT:** When the stem uses an indirect identifier (year instead of name), locate that identifier in the table first.

⚡ **FAST METHOD:** Find "2012" in the year column → read across to mass → 899 → C.

MEDIUM (Q11–Q30)

Q11. Correct Answer: C

REASONING: Calculate the absolute increase for each crop. Rice: 4.9 − 4.5 = 0.4. Wheat: 3.3 − 3.4 = −0.1 (decrease). Maize: 6.7 − 5.9 = 0.8. Sorghum: 1.6 − 1.4 = 0.2. The greatest increase is 0.8 (maize).

WHY EACH OPTION IS WRONG: - A) Rice increased by only 0.4 tonnes, not the greatest. - B) Wheat actually decreased by 0.1 tonnes. - D) Sorghum increased by only 0.2 tonnes.

🚨 **TRAP ALERT:** B is a trap because "greatest increase" and "decrease" are opposite. Some students pick wheat because it changed, without checking the direction. Also, sorghum's percentage increase is high, but the question asks for absolute increase.

💡 **KEY INSIGHT:** "Greatest absolute increase" = largest positive difference, not largest percentage change.

⚡ **FAST METHOD:** Subtract 2018 from 2023 for each crop → largest positive difference → maize (0.8).

Q12. Correct Answer: B

REASONING: Compare optimal pH values: 2.0, 8.0, 6.8, 8.0. Trypsin and lipase both have an optimal pH of 8.0—identical, so the "most similar."

WHY EACH OPTION IS WRONG: - A) Pepsin (2.0) and trypsin (8.0) differ by 6.0. - C) Salivary amylase (6.8) and pepsin (2.0) differ by 4.8. - D) Lipase (8.0) and salivary amylase (6.8) differ by 1.2—closer than A or C, but B is an exact match.

🚨 **TRAP ALERT:** D is the strongest trap because 8.0 and 6.8 are numerically close, but B offers an identical match (8.0 and 8.0).

💡 **KEY INSIGHT:** "Most similar" can mean identical. Check for exact matches before settling on merely close values.

⚡ **FAST METHOD:** Scan the pH column for duplicates → 8.0 appears twice (trypsin and lipase) → B.

Q13. Correct Answer: A

REASONING: Nair's observation is that between any two consecutive data points, the count more than doubled. Check: 120 → 340 (×2.8 ✓), 340 → 1,450 (×4.3 ✓), 1,450 → 4,200 (×2.9 ✓), 4,200 → 9,800 (×2.3 ✓). Option A cites two consecutive transitions that both show more-than-doubling.

WHY EACH OPTION IS WRONG: - B) Simply states 2022 is the highest year—this shows growth but doesn't demonstrate the "more than doubled" pattern between consecutive points. - C) Says the rate of increase "slowed," which contradicts (or at least doesn't support) the observation of consistent more-than-doubling. - D) Gives a total across all years, which is irrelevant to the between-period doubling claim.

TRAP ALERT: C is a trap that introduces an unsupported interpretation ("slowed") using a true data point. The question asks for data that support the observation, not undermine it.

💡 **KEY INSIGHT:** "Support" means the cited data must be consistent with the stated conclusion, and the option must cite specific data, not just restate a trend.

⚡ **FAST METHOD:** The claim is "more than doubled between consecutive points." Check which option cites consecutive pairs showing this. A does; the others don't.

Q14. Correct Answer: A

REASONING: Add percentages for each pair: Niger-Congo + Austronesian = 21.7 + 17.7 = 39.4%. Austronesian + Trans-New Guinea = 17.7 + 6.8 = 24.5%. Trans-New Guinea + Sino-Tibetan = 6.8 + 6.4 = 13.2%. Niger-Congo + Sino-Tibetan = 21.7 + 6.4 = 28.1%. The greatest combined share is 39.4%.

WHY EACH OPTION IS WRONG: - B) 24.5% is less than 39.4%. - C) 13.2% is less than 39.4%. - D) 28.1% is less than 39.4%.

TRAP ALERT: D is tempting because Niger-Congo is the largest single family, but pairing it with the smallest family (Sino-Tibetan) instead of the second-largest (Austronesian) yields a smaller combined total.

💡 **KEY INSIGHT:** "Greatest combined share" = the two largest individual percentages added together.

⚡ **FAST METHOD:** The top two percentages will give the greatest sum. Niger-Congo (21.7) + Austronesian (17.7) = 39.4 → A.

Q15. Correct Answer: D

REASONING: Compare wavelengths of peak absorption: 430, 455, 450, 565. The longest wavelength is 565 nm (phycoerythrin).

WHY EACH OPTION IS WRONG: - A) 430 nm is the shortest wavelength. - B) 455 nm is the second-longest. - C) 450 nm is the third-longest (very close to B).

TRAP ALERT: B (455 nm) and C (450 nm) are close, but neither is the answer. The actual longest wavelength (565 nm) is in a different range entirely.

💡 **KEY INSIGHT:** "Longest wavelength" = largest number in the wavelength column.

⚡ **FAST METHOD:** Scan the wavelength column for the largest number → 565 → phycoerythrin → D.

Q16. Correct Answer: B

REASONING: The researcher's point is that most museums declined but one "bucked the trend." Calculate changes: Louvre −0.7, National Gallery −0.3, Met −1.1, Uffizi +0.7. Only the Uffizi increased. B correctly identifies the Uffizi's increase and notes the others declined.

WHY EACH OPTION IS WRONG: - A) Describes the Louvre's decline but doesn't identify the museum that bucked the trend. - C) Describes the Met's decline—this supports the general trend, not the exception. - D) Gives a 2019 fact about the National Gallery—irrelevant to the "bucked the trend" point.

TRAP ALERT: A and C each cite real data but illustrate the general decline, not the exception. The question asks for the data that illustrates the researcher's specific point about the exception.

💡 **KEY INSIGHT:** When the passage highlights an exception to a trend, the correct answer must identify that exception with supporting data.

⚡ **FAST METHOD:** Which museum went up? Uffizi (4.4 → 5.1). Find that in the options → B.

Q17. Correct Answer: D

REASONING: Herrera's hypothesis: higher water retention → higher crop yield. To weaken this, find a case where higher retention does NOT produce higher yield. Clay has 45% retention but only 3.2 kg/m^2 yield; silt has 38% retention but 4.1 kg/m^2 yield. Higher retention (clay) produced LOWER yield than a soil with less retention (silt). This contradicts the hypothesis.

WHY EACH OPTION IS WRONG: - A) Sand's low retention and low yield support the hypothesis, not weaken it. - C) Sandy loam's higher retention than sand and higher yield support the hypothesis. - B) Silt's higher retention than sandy loam and higher yield support the hypothesis.

TRAP ALERT: Most students instinctively look for data that supports a hypothesis. The question asks for data that WEAKENS it—look for counterexamples.

💡 **KEY INSIGHT:** "Weaken" = find data where the predicted relationship breaks down.

⚡ **FAST METHOD:** Check if the highest water retention (clay, 45%) has the highest yield. It doesn't (silt at 38% has 4.1 vs. clay's 3.2). That's the counterexample → B.

Q18. Correct Answer: B

REASONING: Compare albedo values: 0.16, 0.07, 0.07, 0.06. Hygiea (0.07) and Interamnia (0.07) are identical—the most similar pair.

WHY EACH OPTION IS WRONG: - A) Pallas (0.16) and Hygiea (0.07) differ by 0.09. - C) Interamnia (0.07) and Europa (0.06) differ by 0.01—close but not identical like B. - D) Pallas (0.16) and Europa (0.06) differ by 0.10.

TRAP ALERT: C is the strongest trap. 0.07 and 0.06 are very close (differ by 0.01), but B offers an exact match (both 0.07). "Most similar" favors the identical pair.

💡 **KEY INSIGHT:** Same logic as Q12—check for exact matches before settling on merely close values.

⚡ **FAST METHOD:** Scan albedo column for duplicates → 0.07 appears twice → Hygiea and Interamnia → B.

Q19. Correct Answer: B

REASONING: Okamura's conclusion: doubling the dose didn't produce a meaningful advantage over the standard dose. The data must compare the two dosage groups. Standard dose: 62% relief. Double dose: 64% relief. The difference is only 2 percentage points—not meaningful. B correctly cites these figures.

WHY EACH OPTION IS WRONG: - A) Compares placebo to standard dose, not double dose to standard dose. - C) Compares double dose to placebo, not to standard dose. - D) States sample sizes were equal—this is about study design, not about the comparison between the two dosage levels.

TRAP ALERT: A and C cite real data but compare the wrong groups. The conclusion is specifically about double vs. standard dose.

💡 **KEY INSIGHT:** Match the conclusion's comparison to the data. The conclusion compares two specific groups, so the supporting data must compare those same groups.

⚡ **FAST METHOD:** Conclusion says "doubling ≈ no advantage over standard." Find the option that compares double (64%) to standard (62%) → B.

Q20. Correct Answer: A

REASONING: Calculate percentage-point decrease for each: South America: 51.4 − 46.2 = 5.2. Sub-Saharan Africa: 29.8 − 25.9 = 3.9. Southeast Asia: 49.8 − 46.1 = 3.7. Northern Europe: 54.6 − 53.1 = +1.5 (increase). The largest decrease is 5.2 (South America).

WHY EACH OPTION IS WRONG: - B) 3.9 is a decrease but not the largest. - C) 3.7 is the smallest decrease. - D) Northern Europe increased, so it has no decrease at all.

TRAP ALERT: D is immediately eliminable (it's an increase, not a decrease). The real competition is between A (5.2), B (3.9), and C (3.7).

💡 **KEY INSIGHT:** "Largest percentage-point decrease" requires subtraction for each region, then comparison.

⚡ **FAST METHOD:** Subtract 2020 from 2000 for each region → largest positive result = largest decrease → 5.2 → South America → A.

Q21. Correct Answer: C

REASONING: Osei's claim: geopolymer concrete offers a better balance of lower environmental impact AND structural performance compared to cross-laminated timber. Compare the two: Geopolymer concrete: 0.06 kg CO_2/kg, 28 MPa. Cross-laminated timber: 0.05 kg CO_2/kg, 25 MPa. Geopolymer has slightly higher carbon (0.06 vs. 0.05) but notably higher strength (28 vs. 25 MPa). C captures this trade-off.

WHY EACH OPTION IS WRONG: - A) Compares geopolymer to conventional concrete—not the comparison in the conclusion. - B) States timber has the lowest carbon—true, but this doesn't support the claim that geopolymer is BETTER than timber on balance. - D) Describes recycled steel—irrelevant to the geopolymer vs. timber comparison.

TRAP ALERT: A is tempting because geopolymer does outperform conventional concrete, but the conclusion specifically claims geopolymer is better than cross-laminated timber, not conventional concrete.

💡 **KEY INSIGHT:** The correct answer must compare the same two items the conclusion compares.

⚡ **FAST METHOD:** Conclusion says "geopolymer > timber in balance." Find the option comparing those two → C.

Q22. Correct Answer: C

REASONING: Calculate depth range for each: Spotted goby: 15 − 1 = 14 m. Sand smelt: 20 − 0 = 20 m. Lesser weever: 50 − 1 = 49 m. Rock cook wrasse: 30 − 1 = 29 m. The widest is 49 m (lesser weever).

WHY EACH OPTION IS WRONG: - A) Spotted goby has a range of only 14 m. - B) Sand smelt has a range of 20 m. - D) Rock cook wrasse has a range of 29 m.

TRAP ALERT: D (rock cook wrasse, 29 m range) is the second-widest and could trick students who don't check all rows.

💡 **KEY INSIGHT:** "Widest depth range" = largest difference between maximum and minimum depth, not the deepest maximum.

⚡ **FAST METHOD:** Subtract min from max for each species → 49 is the largest → lesser weever → C.

Q23. Correct Answer: A

REASONING: The two longest novels by word count are War and Peace (~580,000) and Bleak House (~530,000). The student claims they were not published in the same decade. War and Peace: 1869 (1860s). Bleak House: 1853 (1850s). Different decades—the claim holds. A is the only option that correctly identifies both of the two longest novels and provides their publication years.

WHY EACH OPTION IS WRONG: - B) Cites Les Misérables (~360,000) and Middlemarch (~316,000)—the third- and fourth-longest, not the two longest. - C) Cites Bleak House and Middlemarch—only one of the two longest is named. - D) Cites War and Peace and Les Misérables—only one of the two longest (War and Peace is correct, but Les Misérables at ~360,000 is the third-longest, not the second).

TRAP ALERT: D is tempting because it pairs War and Peace (the longest) with Les Misérables from a different year, but Les Misérables (~360,000) is NOT the second-longest. Bleak House (~530,000) is.

💡 **KEY INSIGHT:** First identify which two novels the claim is about (the two longest by word count), then check which option correctly cites both.

⚡ **FAST METHOD:** Rank by word count → top two → War and Peace (580K) + Bleak House (530K) → only A names both → 1869 vs. 1853 confirms different decades.

Q24. Correct Answer: B

REASONING: The homeowner is considering mineral wool as a compromise. The observation should describe mineral wool's position relative to the other options. Mineral wool: R-value 3.3, cost $1.10. B says mineral wool has a higher R-value than fiberglass (3.3 > 3.2 ✓) and costs less than spray foam ($1.10 < $1.50 ✓)—a true compromise statement.

WHY EACH OPTION IS WRONG: - A) Says mineral wool has a lower R-value than cellulose (3.3 < 3.7 ✓) but costs more ($1.10 > $0.82 ✓). While factually correct, this paints mineral wool as worse than cellulose on both metrics—not a compromise argument. - C) Says mineral wool has the "same" R-value as fiberglass —incorrect (3.3 ≠ 3.2). - D) Says mineral wool has the "lowest" R-value—incorrect (fiberglass at 3.2 is lower).

🚨 **TRAP ALERT:** A is factually accurate but doesn't frame mineral wool as a compromise; it makes it look inferior to cellulose. C contains a factual error (3.3 ≠ 3.2).

💡 **KEY INSIGHT:** The answer must serve the rhetorical purpose stated in the stem (framing mineral wool as a compromise), not just state correct data.

⚡ **FAST METHOD:** A "compromise" means mineral wool must beat at least one option on one metric and another on a different metric. B does this.

Q25. Correct Answer: B

REASONING: The student's point: the river with the greatest discharge is NOT the longest. Greatest discharge = Yenisei (19,600 m^3/s). Longest = Ob (5,410 km). These are different rivers, confirming the student's point. B directly states this: the Yenisei has the greatest discharge at 19,600 m^3/s, yet at 3,487 km it is shorter than the Ob, which at 5,410 km is the longest.

WHY EACH OPTION IS WRONG: - A) Describes the Lena, which is neither the longest nor has the greatest discharge—accurate data, but doesn't directly make the student's point about the disconnect between the two leading categories. - C) Describes the Ob as the longest and the Amur as the shortest—true but doesn't identify the river with the greatest discharge (Yenisei), which is the key comparison. - D) Falsely claims the Yenisei is "also the longest river in the table at 5,410 km"—the Yenisei is 3,487 km, and 5,410 km is the Ob's length. This option contains a factual error.

🚨 **TRAP ALERT:** D is a dangerous trap that mislabels the Yenisei's length with the Ob's length (5,410 km). Always verify numbers against the actual table.

💡 **KEY INSIGHT:** The student's claim requires identifying TWO rankings (greatest discharge, longest length) and showing they belong to different rivers.

⚡ **FAST METHOD:** Greatest discharge → Yenisei (19,600). Longest → Ob (5,410 km). Different rivers. Find the option naming both → B.

Q26. Correct Answer: A

REASONING: Calculate boiling point − melting point for each element: Gallium: 2,204 − 29.8 = 2,174.2°C. Cesium: 671 − 28.4 = 642.6°C. Rubidium: 688 − 39.3 = 648.7°C. Indium: 2,072 − 156.6 = 1,915.4°C. Gallium's difference is 2,174.2°C—the largest. A correctly states this and compares it to cesium's 642.6°C.

WHY EACH OPTION IS WRONG: - B) States gallium's boiling point but doesn't calculate the difference (melting point is not mentioned). - C) Says gallium's difference is 1,915.4°C—that's indium's difference, not gallium's. - D) States gallium's melting point—irrelevant to the boiling-melting difference.

🚫 **TRAP ALERT:** C swaps gallium's difference with indium's. Always recalculate rather than trusting the option's arithmetic.

💡 **KEY INSIGHT:** When an option presents a calculation, verify it independently against the table.

⚡ **FAST METHOD:** Subtract melting from boiling for gallium: 2,204 − 29.8 = 2,174.2. Match to options → A.

Q27. Correct Answer: A

REASONING: Beaulieu's conclusion: native patches attract a "wider range of frequent pollinators." This means more species visit frequently (not just one dominant species). In the wildflower patch: honeybee 12, bumblebee 18, hoverfly 8, sweat bee 14. Three species exceed 10 visits/hr. In the ornamental garden: honeybee 15, bumblebee 7, hoverfly 9, sweat bee 4. Only one species (honeybee) exceeds 10 visits/hr. A captures this contrast.

WHY EACH OPTION IS WRONG: - B) Shows hoverflies visit the ornamental garden slightly more (9 vs. 8)—this weakens the conclusion and addresses only one species, not the "wider range." - C) Shows honeybees prefer ornamental gardens—doesn't address "wider range." - D) Describes one species in one habitat—doesn't make the comparative argument.

🚫 **TRAP ALERT:** B and C cite real data but support the opposite conclusion (that ornamental gardens attract some pollinators more). D is too narrow—one data point doesn't establish "wider range."

💡 **KEY INSIGHT:** "Wider range of frequent pollinators" requires showing that MORE species are active in one habitat vs. the other.

⚡ **FAST METHOD:** Count species with high visit rates in each habitat. Wildflower: 3 species above 10. Ornamental: 1 species above 10. Only A makes this comparison → A.

Q28. Correct Answer: B

REASONING: The researcher's point: reducing water ≠ reducing carbon. Need a case where water is low but carbon is high. Polyester: 17 liters/kg (by far the lowest water usage) but 9.5 kg CO_2/kg (the highest carbon). This perfectly illustrates the disconnect.

WHY EACH OPTION IS WRONG: - A) Organic cotton has lower water AND lower carbon than conventional cotton—this shows the two metrics moving TOGETHER, the opposite of the researcher's point. - C) Linen has lower water AND lower carbon than conventional cotton—same problem as A. - D) Describes conventional cotton's high values on both metrics—again shows correlation, not disconnect.

🚫 **TRAP ALERT:** A, C, and D all show cases where water and carbon move in the same direction. The question asks for a case where they move in opposite directions.

💡 **KEY INSIGHT:** "Trade-off" or "disconnect" questions need a counterexample—data where one metric improves while the other worsens.

⚡ **FAST METHOD:** Find lowest water → polyester (17). Check its carbon → 9.5 (highest!). That's the disconnect → B.

Q29. Correct Answer: A

REASONING: Each region has a dominant term: Upper Midwest = "pop" (82%), Northeast = "soda" (75%), Southeast = "coke" (68%), Pacific Northwest = "soda" (52%). The question asks which region's dominant term has the highest usage rate across ALL regions. "Pop" in the Upper Midwest is at 82%—the highest single percentage in the entire table.

WHY EACH OPTION IS WRONG: - B) "Soda" in the Northeast is at 75%—high but lower than 82%. - C) "Coke" in the Southeast is at 68%—lower than 82%. - D) "Soda" in the Pacific Northwest is at 52%—the lowest dominant-term percentage.

TRAP ALERT: The question has two parts: (1) each region has a dominant term, and (2) which region's dominant percentage is the highest overall. Students who miss part (2) might pick any region.

💡 **KEY INSIGHT:** Scan the entire table for the single highest percentage in any cell → 82%.

⚡ **FAST METHOD:** Highest percentage anywhere in the table → 82% → Upper Midwest, "pop" → A.

Q30. Correct Answer: D

REASONING: Shortest orbital period: Proxima Centauri b at 11 days. Highest surface temperature: TOI-700d at 5°C. These are different planets, confirming the astrobiologist's observation. A correctly identifies both.

WHY EACH OPTION IS WRONG: - B) Describes Kepler-442b but doesn't address the shortest orbital period or highest temperature. - C) Falsely claims TOI-700d has the shortest orbital period (it has 37 days; Proxima Centauri b has 11). - A) Describes Kepler-186f but doesn't address the core observation about shortest period vs. highest temperature.

TRAP ALERT: C is a dangerous trap—it gets the temperature right (TOI-700d is warmest) but falsely claims TOI-700d has the shortest orbital period. Always verify ALL claims in an option.

💡 **KEY INSIGHT:** Two separate lookups are needed: (1) find the shortest orbital period, (2) find the highest temperature, then confirm they're different planets.

⚡ **FAST METHOD:** Shortest period → Proxima Centauri b (11 days). Highest temp → TOI-700d (5°C). Different planets. Which option says this? → A.

HARD (Q31–Q50)

Q31. Correct Answer: A

REASONING: Watanabe's conclusion: amendments help most under drought (big gains) but add little when water is plentiful (marginal gains). A compares the biochar + mycorrhizal fungi treatment across conditions: Under drought: 16.1 − 8.2 = 7.9 cm gain over control. Under irrigation: 16.0 − 14.6 = 1.4 cm gain over control. The drought gain is 5.6× larger, perfectly illustrating the conclusion.

WHY EACH OPTION IS WRONG: - B) Cites mycorrhizal fungi's irrigated performance but doesn't compare to drought or to the control under both conditions. - C) Notes that the biochar + mycorrhizal treatment produces similar absolute root lengths across conditions (16.1 vs. 16.0), but this describes the treatment's consistency, not its advantage over the control. - D) Shows biochar's gains in both conditions but doesn't explicitly highlight the disparity in gain magnitude.

TRAP ALERT: C is the strongest trap. It's true that the combined treatment produces ~16 cm in both conditions, but the conclusion is about gain RELATIVE TO CONTROL, not absolute performance. The control under drought is 8.2 (big gap), while under irrigation it's 14.6 (small gap).

💡 **KEY INSIGHT:** "Benefit is greatest under X" means the DIFFERENCE between treatment and control should be larger under X. You must compare differences, not just treatment values.

⚡ **FAST METHOD:** Calculate treatment − control under drought vs. under irrigation. Largest disparity in differences → A.

Q32. Correct Answer: D

REASONING: Moreau's conclusion has two parts: (1) dominant grasses recovered well, (2) forb species remain underrepresented. Grasses: Prairie dropseed (0.42 vs. 0.45 expected, gap = 0.03) and big bluestem (0.28 vs. 0.30, gap = 0.02)—close to expected. Forbs: Wild bergamot (0.05 vs. 0.12, gap = 0.07) and pale purple coneflower (0.03 vs. 0.10, gap = 0.07)—far below expected. A captures both halves.

WHY EACH OPTION IS WRONG: - B) Claims "all four species" are below expected—true, but it misses the key distinction between grasses (slightly below) and forbs (substantially below). The conclusion is about that contrast. - C) Only discusses prairie dropseed, not the contrast between grasses and forbs. - A) Incorrectly claims the grass species "exceed their expected frequencies"—they don't ($0.42 < 0.45$ and $0.28 < 0.30$).

🚫 **TRAP ALERT:** D contains a factual error (grasses don't exceed expected values). B is true but fails to highlight the contrast, which is the core of the conclusion.

💡 **KEY INSIGHT:** When a conclusion draws a contrast between two categories, the correct answer must address BOTH categories.

⚡ **FAST METHOD:** Conclusion = "grasses OK, forbs not OK." Find the option that shows both sides → A.

Q33. Correct Answer: B

REASONING: Solheim's conclusion: (1) CO_2 and methane generally increased together, and (2) the most substantial joint increase was in the most recent transition. Check transitions going forward in time: D→C: CO_2 +5, CH_4 +10. C→B: CO_2 +30, CH_4 +60. B→A: CO_2 +50, CH_4 +230. The B→A transition (the most recent) has the largest increases in both gases. B addresses both parts: it shows the overall C-to-A rise and then highlights the B→A transition as the single largest joint increase.

WHY EACH OPTION IS WRONG: - A) Describes the D→C transition as having the smallest increases—true but doesn't support the conclusion about the most recent transition being the most substantial. - C) Only notes that Layer A has the highest values—true, but doesn't describe transitions or identify the most recent one as the largest. - D) Describes C→B and D→C transitions and claims the rate accelerated—partially true but doesn't identify B→A (the most recent) as the most substantial. It also doesn't directly address the conclusion's emphasis on the "most recent" transition.

🚫 **TRAP ALERT:** C is tempting because it confirms the endpoint, but the conclusion is about the pattern of change (transitions), not just final values. D partially supports accelerating increases but focuses on older transitions rather than the most recent one.

💡 **KEY INSIGHT:** When a conclusion involves trends over time, the correct answer should describe changes between data points and specifically identify the transition the conclusion highlights.

⚡ **FAST METHOD:** Calculate each transition: D→C (+5, +10), C→B (+30, +60), B→A (+50, +230). B→A is clearly the largest. B is the only option citing this. → B.

Q34. Correct Answer: A

REASONING: El-Amin's conclusion: (1) all treatments reduced counts vs. control, (2) silver was most effective, achieving several orders of magnitude reduction by Day 7. Silver: $5.0 \times 10^6 \rightarrow 1.5 \times 10^3$ = reduction by ~3,300× (more than 3 orders of magnitude). Control: grew to 1.2×10^7. A cites both the silver reduction and the control increase.

WHY EACH OPTION IS WRONG: - B) Describes Day 3 data and zinc oxide's limited effectiveness—partially relevant but doesn't quantify silver's Day 7 superiority. - C) Describes copper—about one order of magnitude reduction, less dramatic than silver's. - D) Only describes the control's growth—doesn't address any treatment's effectiveness.

🚫 **TRAP ALERT:** B and C describe real data about other treatments but don't address the core conclusion about silver's unique potency. D is necessary context but insufficient alone.

💡 **KEY INSIGHT:** "Most effective" conclusions require data showing the best performer's specific results AND a reference point (control or weaker treatment) for comparison.

⚡ **FAST METHOD:** Silver Day 7 = 1.5×10^3 vs. initial 5.0×10^6 = >3 orders of magnitude drop. Find this in options → A.

Q35. Correct Answer: D

REASONING: Strand's conclusion: distractions increase response times overall, but the increase is

DISPROPORTIONATELY larger for incongruent trials. This means the gap between incongruent and congruent should widen as distractions increase. No distraction: 420 − 340 = 80 ms gap. Dual distraction: 610 − 410 = 200 ms gap. The gap more than doubles (80 → 200), showing a disproportionate effect on incongruent trials. A captures this.

WHY EACH OPTION IS WRONG: - B) Cites ranges and a single maximum but doesn't calculate or compare the gap between congruent and incongruent across conditions. - C) Compares visual vs. auditory distraction—a different question from the one about proportionality of effect on congruent vs. incongruent trials. - A) Notes that dual distraction has the highest times for both—this shows "more distractions = slower" but doesn't demonstrate the disproportionate effect on incongruent trials specifically.

TRAP ALERT: D is the strongest trap—it's true and seems to support the conclusion, but it shows EQUAL effects on both trial types, not disproportionate ones. The key word in the conclusion is "disproportionately."

💡 **KEY INSIGHT:** "Disproportionate" means the GAP between two conditions changes, not just that both increase. Calculate the difference between incongruent and congruent response times across distraction levels.

⚡ **FAST METHOD:** Gap in no-distraction: 420 − 340 = 80. Gap in dual distraction: 610 − 410 = 200. Gap grows → disproportionate → A.

Q36. Correct Answer: D

REASONING: Banerjee's conclusion: (1) satisfaction generally increases with GDP, but (2) the relationship isn't perfectly linear—some poorer countries outscore wealthier peers. B identifies TWO exceptions: Country B ($48K, 7.6) > Country A ($82K, 7.4)—a less wealthy country is more satisfied. Country E ($3.2K, 6.3) > Country D ($5.5K, 5.8)—same pattern. These exceptions demonstrate the non-linearity.

WHY EACH OPTION IS WRONG: - A) Shows the wealthiest country has high satisfaction and the poorest has low satisfaction—this supports a linear relationship, the opposite of the conclusion about non-linearity. - C) Groups three low-income countries together but doesn't show any exception to the pattern. - B) Shows the same linear pattern as A (high GDP = high satisfaction, low GDP = low satisfaction).

TRAP ALERT: A and D both show data consistent with a LINEAR relationship. The conclusion's key point is the NON-linearity—exceptions where lower income → higher satisfaction.

💡 **KEY INSIGHT:** A conclusion about a non-linear or imperfect relationship requires data showing exceptions or reversals, not confirmations.

⚡ **FAST METHOD:** Find pairs where lower GDP → higher satisfaction. B ($48K > $82K in GDP, but 7.6 > 7.4 in satisfaction) and E ($3.2K < $5.5K, but 6.3 > 5.8). Only B identifies these → B.

Q37. Correct Answer: A

REASONING: Lindgren's conclusion has two parts: (1) epoxy has the highest initial strength, (2) polyurethane is best for prolonged submersion due to the most stable bond. Verify stability: Epoxy drops 8.7 MPa (28.5 → 19.8). Cyanoacrylate drops 18.8 MPa (22.0 → 3.2). Polyurethane drops only 1.3 MPa (15.4 → 14.1). Marine-grade silicone drops 3.4 MPa (8.2 → 4.8). Polyurethane has the smallest decline—it IS the most stable. A compares epoxy's large decline to polyurethane's small decline, directly supporting the conclusion.

WHY EACH OPTION IS WRONG: - B) Describes marine-grade silicone's large decline (over 40%)—this supports the background context but doesn't compare polyurethane to the initial strength leader (epoxy). - C) Describes cyanoacrylate's failure—supports part of the background but doesn't directly support the claim about polyurethane. - D) Contains a false comparison: it says polyurethane's 72-hr strength (14.1) is higher than cyanoacrylate's dry strength (22.0)—but 14.1 < 22.0.

TRAP ALERT: D is dangerous because students may not catch the false arithmetic (14.1 is NOT higher than 22.0). Always verify numerical comparisons in options against the table.

💡 **KEY INSIGHT:** When a conclusion names a specific winner, the supporting data should compare that winner against the relevant alternative mentioned in the conclusion (epoxy, the initial strength leader).

⚡ **FAST METHOD:** Conclusion: "epoxy = highest initial, polyurethane = most stable." Compare their declines: epoxy −8.7 MPa vs. polyurethane −1.3 MPa. Only A makes this comparison → A.

Q38. Correct Answer: B

REASONING: Chen's conclusion: green space doesn't consistently track with income; some lower-income neighborhoods have more green space. B identifies Riverside ($55K income, 52 m^2 green space) exceeding both Westbrook ($92K, 45 m^2) and Northgate ($78K, 30 m^2) despite lower income. This directly demonstrates the disconnect.

WHY EACH OPTION IS WRONG: - A) Compares the top two income neighborhoods—both have income and green space roughly aligned (higher income → more green space), supporting a correlation rather than a disconnect. - C) Shows Elm Terrace with lower income AND less green space—this supports correlation, not the disconnect Chen claims. - D) Discusses commute time, which is irrelevant to the income-vs-green-space conclusion.

🚨 **TRAP ALERT:** A and C both show data where income and green space correlate positively—the opposite of what the conclusion claims. D introduces an irrelevant variable.

💡 **KEY INSIGHT:** "Does not consistently track" = find counterexamples where the expected relationship breaks down.

⚡ **FAST METHOD:** Find a low-income neighborhood with high green space. Riverside: low income ($55K), highest green space (52 m^2). Find this in options → B.

Q39. Correct Answer: A

REASONING: Park's conclusion: the most complex model (neural network) doesn't offer the best TRADE-OFF between performance on new data and computational efficiency. A shows: neural network gains only 2 percentage points over random forest on new data (86% vs. 84%) but requires 5.3× the processing time (45.0 vs. 8.5 seconds). The marginal accuracy gain doesn't justify the massive computational cost.

WHY EACH OPTION IS WRONG: - B) Describes the decision tree's overfitting—relevant to model evaluation but doesn't address the trade-off claim about the most complex model. - C) Describes linear regression—the simplest, not the most complex model. - D) Claims random forest is the "overall best"—this is an unsupported conclusion, and the question asks about the neural network's trade-off.

🚨 **TRAP ALERT:** D sounds appealing because random forest does perform well, but the conclusion is specifically about the neural network's poor trade-off, not about which model is best overall. The correct answer must address the neural network.

💡 **KEY INSIGHT:** "Trade-off" arguments require comparing the marginal benefit (accuracy gain) against the marginal cost (processing time). The correct answer must quantify both sides.

⚡ **FAST METHOD:** Neural network vs. random forest: +2% accuracy but +36.5 seconds. Poor trade-off. Only A shows this → A.

Q40. Correct Answer: D

REASONING: Fernandez's conclusion: (1) organic amendments start lower than synthetic but improve over time, and (2) this is likely because they build soil nitrogen. A addresses both: Synthetic NPK starts highest (52) but declines (48 by Year 3). Compost starts lower (38) but rises to 50, surpassing synthetic by Year 3. Compost's soil nitrogen (28 mg/kg) is more than double synthetic's (12 mg/kg), supporting the nitrogen-building mechanism.

WHY EACH OPTION IS WRONG: - B) Claims organic amendments are "consistently inferior"—false by Year 3 (compost 50 > synthetic 48). - C) Describes the no-fertilizer control—relevant background but doesn't compare organic to synthetic, which is the core of the conclusion. - A) Notes Year 3 values are close but doesn't show the trajectory (initially lower, then rising) or the soil nitrogen mechanism.

TRAP ALERT: D presents accurate Year 3 data but misses the conclusion's emphasis on the TRAJECTORY (initially lower → improving) and the nitrogen mechanism. The conclusion is about a trend, not a snapshot.

💡 **KEY INSIGHT:** When a conclusion involves a temporal trend AND a causal mechanism, the correct answer must address both elements.

⚡ **FAST METHOD:** Two claims: (1) organic starts low, catches up → compost 38→50 vs. synthetic 52→48. (2) Nitrogen reason → compost 28 vs. synthetic 12 mg/kg. Only A includes both → A.

Q41. Correct Answer: B

REASONING: Andreou's conclusion: trade peaked in 100–300 CE and subsequently declined, indicating both volume and diversity peaked then dropped. B shows 100–300 CE leads all three metrics: most shipwrecks (112), largest cargo (140 tonnes), highest ceramics percentage (62%), AND notes all three declined afterward. This comprehensively supports both "peaked" and "declined."

WHY EACH OPTION IS WRONG: - A) Shows the increase before the peak—supports the "rise" but not the "peak and decline." - C) Shows the decline after 300 CE—supports only the declining half, not the peak itself. - D) Notes the ceramics decline but suggests an alternative interpretation (changing preferences), which undermines rather than supports the trade-contraction conclusion.

TRAP ALERT: C is tempting because it shows the decline, but it only covers half the conclusion. The correct answer must show both the peak (why 100–300 CE is the highest) AND the decline.

💡 **KEY INSIGHT:** When a conclusion involves "peaked and then declined," the correct answer must establish the peak values AND show subsequent decreases.

⚡ **FAST METHOD:** The period 100–300 CE must lead all metrics AND subsequent periods must show declines. Only B confirms both → B.

Q42. Correct Answer: A

REASONING: Brooks's conclusion: (1) urban environments constrain all species, (2) the magnitude varies among species. "Magnitude" → proportional reduction. White oak: urban 15.2 / rural 42.3 = 36% of rural size (64% reduction). American sycamore: urban 22.6 / rural 38.9 = 58% of rural size (42% reduction). White oak is proportionally more reduced than sycamore. A captures this contrast.

WHY EACH OPTION IS WRONG: - B) Says urbanization constrains growth "uniformly"—this contradicts the conclusion that magnitude varies among species. - C) States sycamore has the largest urban DBH—true, but doesn't address proportional reduction relative to rural values. - D) Notes hemlock's low urban DBH—true, but doesn't compare proportional reductions across species.

TRAP ALERT: B directly contradicts the conclusion ("uniformly" vs. "varies among species"). C and D each focus on absolute values rather than proportional reductions.

💡 **KEY INSIGHT:** "Magnitude varies" requires comparing proportional (not absolute) changes across species. Calculate urban/rural ratios.

⚡ **FAST METHOD:** White oak: 15.2/42.3 ≈ 36%. Sycamore: 22.6/38.9 ≈ 58%. Different proportional reductions → illustrates variable magnitude → A.

Q43. Correct Answer: D

REASONING: Adeyemi's conclusion: Method A is better for low-performing students; Method B is better for high-performing students. A provides both comparisons: Low-performing: Method A gain = 58 − 32 = 26 pts. Method B gain = 49 − 32 = 17 pts. Method A wins by 9 pts. High-performing: Method A gain = 85 − 78 = 7 pts. Method B gain = 88 − 78 = 10 pts. Method B wins by 3 pts. Both halves of the conclusion are supported.

WHY EACH OPTION IS WRONG: - B) Shows Method B beats Method A for mid-performers—partially relevant but doesn't address the two groups named in the conclusion (low and high). - C) Notes low-performers gain the most overall—true but doesn't compare Method A to Method B. - A) Notes the small high-performer difference might not be significant—this WEAKENS the conclusion rather than supporting it.

🚨 **TRAP ALERT:** D is a trap that introduces doubt about statistical significance, effectively undermining one half of the conclusion. B addresses the wrong student tier (mid, not low or high).

💡 **KEY INSIGHT:** When a conclusion makes claims about TWO specific subgroups, the correct answer must provide comparison data for BOTH subgroups.

⚡ **FAST METHOD:** Low: Method A (+26) > Method B (+17). High: Method B (+10) > Method A (+7). Only A shows both → A.

Q44. Correct Answer: D

REASONING: Kováčová's conclusion: premium catalysts (Pt, Pd) show diminishing marginal returns in rate improvement relative to their selectivity advantage. A shows: Pt rate roughly doubles (0.042 → 0.089, ×2.1) while Ni nearly quadruples (0.015 → 0.058, ×3.9). So at higher temperatures, the rate gap between premium and non-premium catalysts shrinks—Pt's rate advantage diminishes. But Pt's selectivity (92%) remains far above Ni's (74%)—the selectivity gap persists. This illustrates "diminishing marginal returns in rate" relative to a persistent "selectivity advantage."

WHY EACH OPTION IS WRONG: - B) Describes iron as the worst catalyst—accurate but doesn't address the premium-vs-non-premium comparison or the concept of diminishing returns. - C) States that temperature helps all catalysts—true but too general. Doesn't address the differential rate improvement or selectivity. - A) Compares Pd to Pt—both are premium catalysts. The conclusion contrasts premium vs. non-premium.

🚨 **TRAP ALERT:** D is tempting because it compares two named catalysts, but the conclusion is about premium vs. non-premium. The relevant comparison is Pt (or Pd) vs. Ni (or Fe).

💡 **KEY INSIGHT:** "Diminishing marginal returns" means the advantage gets smaller as conditions change. Calculate the rate multiplier for premium vs. non-premium at both temperatures.

⚡ **FAST METHOD:** Pt rate increase: ×2.1. Ni rate increase: ×3.9. Non-premium closes the gap → Pt's rate advantage diminishes → but selectivity gap stays. A captures this → A.

Q45. Correct Answer: B

REASONING: Boulos's conclusion: documentation intensified but extinction accelerated EVEN FASTER, so documentation alone is insufficient. B compares growth rates: Extinct languages: 12 → 62 = more than fivefold increase. Written grammars: 85 → 130 = less than doubling. Extinction grew much faster than documentation, supporting the claim that documentation efforts haven't kept pace.

WHY EACH OPTION IS WRONG: - A) Notes that written grammars peaked in the 1990s and declined—relevant but doesn't compare documentation growth to extinction growth. - C) Describes audio documentation growth—this shows documentation intensifying but doesn't compare to extinction rates. - D) Gives 2000s data for both metrics but doesn't compare growth rates over time—it's a snapshot, not a trend comparison.

🚨 **TRAP ALERT:** A is a strong trap because it shows documentation declining, which supports "insufficient," but it doesn't compare growth rates between documentation and extinction—the core of the conclusion.

💡 **KEY INSIGHT:** "X accelerated faster than Y" requires comparing the RATE OF GROWTH of both X and Y across the same period.

⚡ **FAST METHOD:** Extinction multiplier: 62/12 ≈ 5.2×. Documentation multiplier: 130/85 ≈ 1.5×. Extinction grew 3.5× faster. Only B makes this comparison → B.

Q46. Correct Answer: C

REASONING: Delgado's conclusion: drip methods offer (1) comparable or higher yields, (2) dramatically lower water use, and (3) reduced salinization. A provides all three comparisons: Drip irrigation: 6.5 t/ha (highest yield), 3,100 L/day (much less than flood's 8,500), salinity 1.9 dS/m (much less than flood's 4.8). This comprehensively supports all three parts of the conclusion.

WHY EACH OPTION IS WRONG: - B) Compares subsurface drip to drip irrigation—both are drip-based methods. The conclusion contrasts drip methods vs. conventional methods. - A) Describes sprinkler irrigation—not a drip method, so it's not the subject of the conclusion. - D) Only describes flood irrigation —doesn't compare to drip methods.

🚨 **TRAP ALERT:** B is a subtle trap. It's comparing two drip systems to each other, but the conclusion is about drip vs. conventional (flood/sprinkler). The comparison group matters.

💡 **KEY INSIGHT:** When a conclusion contrasts two categories, the supporting data must compare items from different categories, not within the same category.

⚡ **FAST METHOD:** Conclusion: "drip > conventional." Compare drip (3,100 L, 6.5 t/ha, 1.9 dS/m) to flood (8,500 L, 6.2 t/ha, 4.8 dS/m) → drip wins on all three → A.

Q47. Correct Answer: A

REASONING: Reyes's conclusion: disturbance reduces (1) species richness, (2) evenness, and (3) increases invasives. A compares the two extremes: Old-growth (least disturbed): 42 species, 0.85 evenness, 3% invasive. Plantation (most disturbed): 18 species, 0.45 evenness, 22% invasive. All three metrics move in the direction predicted by the conclusion.

WHY EACH OPTION IS WRONG: - B) States urban parks have higher richness but LOWER evenness than plantations—but this is wrong: urban parks actually have HIGHER evenness (0.60 > 0.45). B contains a factual error. - C) Describes secondary forests' partial recovery—relevant but only addresses one habitat, not the contrast between disturbed and undisturbed. - D) Acknowledges the invasive trend but suggests an alternative explanation (deliberate planting)—this weakens rather than supports the conclusion.

🚨 **TRAP ALERT:** D cites accurate data but then offers a competing explanation that undermines the conclusion. Data + alternative interpretation ≠ support.

💡 **KEY INSIGHT:** When a conclusion involves three parallel trends, the best supporting evidence shows all three moving in the predicted direction across the same contrast.

⚡ **FAST METHOD:** Compare the extremes: old-growth (best on all 3 metrics) vs. plantation (worst on all 3). Only A makes this full comparison → A.

Q48. Correct Answer: B

REASONING: Voronov's conclusion has two parts: (1) social media displaced print, AND (2) the landscape is fragmenting (multiple platforms growing, not just one replacing another). B addresses both: Social media 15% → 42%, print 42% → 9% (displacement ✓). But also podcasts 8% → 23% (nearly tripled), while online news 35% → 26% (declining). Multiple platforms shifting in different directions = fragmentation, not simple migration. B captures the full picture.

WHY EACH OPTION IS WRONG: - A) Only shows the print-to-social-media shift—supports part (1) but describes a "simple transfer," directly contradicting the fragmentation claim in part (2). - C) Claims digital consumption is declining—false (social media + podcasts are growing; only online news sites declined). - D) Combines social media + podcasts to show digital dominance—true, but doesn't illustrate the fragmentation across multiple platforms.

🚨 **TRAP ALERT:** A is the most dangerous trap. It accurately describes the print-to-social-media shift, but the conclusion specifically says the landscape is "fragmenting rather than simply migrating from one platform to another." A describes exactly the simple migration the conclusion rejects.

💡 **KEY INSIGHT:** When a conclusion contains a "not just X, but also Y" structure, the correct answer must address both X and Y. An option that only addresses X (even if perfectly accurate) is insufficient.

⚡ **FAST METHOD:** Conclusion: "displacement + fragmentation." A = displacement only. B = displacement + fragmentation. C = false claim. D = aggregation, not fragmentation → B.

Q49. Correct Answer: D

REASONING: Montague's conclusion: sunlight exposure during working hours predicts vitamin D better than age. A provides a multi-group comparison: Groups with limited daytime outdoor exposure (night-shift nurses: 18.6 ng/mL, 52% deficient; office workers: 22.4 ng/mL, 38% deficient) have lower levels and higher deficiency than outdoor laborers (34.7 ng/mL, 9% deficient). Crucially, A notes that these indoor workers are younger than retired adults (65+), who nonetheless have better vitamin D status (25.1 ng/mL, 31% deficient). This shows that occupation (sunlight exposure) is more predictive than age.

WHY EACH OPTION IS WRONG: - B) Only describes night-shift nurses—too narrow, and "nighttime work is harmful to health" is an overbroad conclusion not in the passage. - C) Focuses on variability (standard deviation) among retirees—doesn't compare occupational groups or address the exposure-vs-age claim. - A) Makes a point about distributions vs. means—interesting statistics but doesn't address the core exposure-vs-age comparison.

🚨 **TRAP ALERT:** B is appealing because night-shift nurses do have the worst values, but it doesn't compare across groups or address the age-vs-exposure contrast. C introduces a statistical concept (variability) not relevant to the conclusion.

💡 **KEY INSIGHT:** "X is a stronger predictor than Y" requires showing groups that differ on X (exposure) but are similar on Y (age) still show different outcomes—or vice versa. A does this by noting that younger indoor workers fare worse than older retirees.

⚡ **FAST METHOD:** The age-vs-exposure test: younger indoor workers (nurses, 52% deficient) vs. older retirees (31% deficient). If age were the primary driver, retirees should be worse. They're not → exposure matters more → A.

Q50. Correct Answer: C

REASONING: Gupta's conclusion: the relationship between emissions and ecological impact is nonlinear, with disproportionately large ecological damage in the mid-range of warming. Calculate reef risk change per °C for each transition: SSP1→SSP2: 25 pts / 1.0°C = 25 pts/°C. SSP2→SSP3: 40 pts / 1.0°C = 40 pts/°C. SSP3→SSP5: 23 pts / 1.0°C = 23 pts/°C (ceiling effect near 100%). The SSP2→SSP3 transition shows a markedly higher rate (40 pts/°C) than the SSP1→SSP2 transition (25 pts/°C), demonstrating that in the critical mid-range, each degree of warming produces a disproportionately greater impact. C cites exactly this comparison.

WHY EACH OPTION IS WRONG: - A) Shows that the SSP1→SSP2 and SSP3→SSP5 transitions have similar rates (25 vs. 23 pts per °C), suggesting a consistent rather than accelerating relationship—this fails to capture the mid-range spike. - B) Shows the endpoints but doesn't address nonlinearity—the data is consistent with a linear interpretation too. - D) Describes sea level rise as roughly proportional to temperature—this demonstrates linearity in sea level, contradicting the nonlinearity claim about reef risk.

🚨 **TRAP ALERT:** A is the most dangerous trap. It compares the first and last transitions, which happen to be similar (25 vs. 23 pts/°C), creating the false impression of linearity. But it skips the critical middle transition (40 pts/°C) where the nonlinearity is most apparent.

💡 **KEY INSIGHT:** When evaluating nonlinearity claims, examine all transitions, not just the endpoints. The disproportionate impact may be concentrated in a specific range.

⚡ **FAST METHOD:** Calculate pts/°C for each step: 25 → 40 → 23. The spike at SSP2→SSP3 (40 pts/°C) demonstrates the nonlinearity. Only C highlights this mid-range disproportionality → C.

TOPIC SUMMARY

🔴 TOP 3 MISTAKES STUDENTS MAKE

1. **Wrong-row / wrong-column errors.** On easy and medium questions, the most common mistake is reading data from an adjacent row or the wrong column. When a table has multiple numeric columns (e.g., two years or two conditions), students often pull from the wrong one. Fix: trace your finger from the keyword in the stem across to the correct column header.
2. **Choosing data that is accurate but doesn't match the conclusion.** On medium and hard questions, every option typically contains real data from the table. The trap is selecting data that is true but addresses the wrong claim—for example, data supporting a general trend when the conclusion is about an exception, or data comparing the wrong pair of groups. Fix: restate the conclusion in your own words before evaluating options, and check that the option's comparison matches the conclusion's comparison.
3. **Confusing absolute values with relative changes.** Many conclusions involve comparisons of gains, proportions, or trade-offs. Students pick options that cite the largest absolute number rather than the largest difference, fastest growth rate, or most relevant contrast. Fix: when the conclusion is about a change, trend, or trade-off, calculate differences or ratios rather than looking at raw values.

🟢 TOP 3 RULES TO REMEMBER

1. **Match the comparison.** If the conclusion compares Group A to Group B, the correct answer must compare Group A to Group B—not Group A to Group C, and not just describe Group A in isolation.
2. **Support ≠ "true."** All four options in data questions usually contain factually correct data. "Support" means the data must specifically corroborate the stated conclusion—same groups, same direction, same relationship. An option can be 100% true and still be wrong if it doesn't address the conclusion.
3. **Read the full option before selecting.** Hard data questions often have options that start accurately but end with a false claim, a reversed relationship, or an alternative interpretation that undermines the conclusion. Read every word.

📊 EXPECTED FREQUENCY PER MODULE

Evidence — Data/Graph/Table questions typically appear **2–4 times per module** on the Digital SAT Reading and Writing section. Across both modules, expect **4–7 total questions** per test. Easy versions (simple lookup) appear in Module 1 and early Module 2. Medium and hard versions (supporting conclusions, illustrating trends, comparing data across conditions) appear in Module 2, especially the hard-adaptive pathway. This question type has been among the most consistent in frequency across the 2024, 2025, and 2026 administrations.

END OF DOC 2 — 50 SOLUTIONS